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Measuring Agricultural Spray Droplet Distributions in Propeller Wake: A Cautionary Tale

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The effect of propeller wake, nozzle locations, and the number of propellers on agricultural spray droplet distribution was experimentally investigated through several methods. Standard agricultural nozzles were tested at varying transverse and downstream distances from a pair of APC 17x7 propellers at different RPMs. Droplet size distributions were measured by traversing the nozzle and the propeller across a laser diffraction instrument. The droplet size distribution in the wake indicated a shift in the droplet distribution to larger droplet sizes. However, gutter tests, shadowgraph images, and an extinction-based post processing technique, show that the propeller wake causes the spray plume to widen and induce oscillations to the spray pattern. This leads to a significant decrease in the probability of droplets in the spray due to the finer droplets being carried away by the wake before they reach the laser diffraction measurement zone. As such, we show that conventional laser diffraction measurement results in the wake of a propeller are inherently biased and cannot be trusted as the sole means of testing to estimate drift potential of a drone spray. A secondary methodology is necessary to capture the entire wake effects on spray.

I. Nomenclature

V_e	Mean propeller wake axial velocity (m/s)
W	Weight of the drone (kg)
n	Number of propellers
ρ	Fluid density (kg/m3)
A	Rotor disk area (m2)
V_0	Rotor inflow velocity (m/s)
D_{V_X}	Volume diameter with X% of liquid atomized into droplets below this value
R_{prop}	Radius of the propeller(s) used. R_{Prop} = 17”
RMS	Root mean squared error
I_{mean}	Mean intensity
$I_{Instant}$	Instantaneous intensity
n	Number of images

II. Introduction

In recent years the use of unmanned aerial vehicles (UAVs) for agricultural spraying has increased exponentially especially in Asian countries. As of 2020 there were nearly 110,000 UAVs being used for agricultural spraying with

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an operational area of over 60 million hectares in China alone [1], while in Japan more than 50% of rice paddies are treated by UAVs [2]. The increase of UAVs for agricultural spraying is due to the many advantages they provide compared to the traditional methods of spraying like backpack spraying, tractor boom spraying, and fixed wing aerial aircraft spraying. These traditional methods can be inefficient, labor intensive, and pose a risk to human health, while also being expensive and limited by topography and canopy height [3, 4, 5]. UAVs on the other hand, are relatively cheap and readily available. They are portable, offer virtually limitless environmental adaptability, and don't require a runway. While the use of UAVs for agricultural spraying has many advantages, it may increase the risk of off target contamination due to spray drift. While drone usage has increased, the spray drift potential using UAVs has not been successfully quantified. Drone spraying has taken a trial-and-error approach due to limited and sometimes conflicting research studies on the influence of propeller wake in droplet distribution from an agricultural nozzle in hover and forward flight conditions.

Most UAVs used in agricultural spraying use a multi-rotor configuration with 4 or more rotors. The specifications for some commonly used agricultural UAVs are listed in Table 1. The weight of the agricultural UAVs ranges from 55 to 169 lbs and mostly have either four or six rotors. The propeller diameter ranges from 9" to 20".

Table 1 Specifications of DJI Agridrones [6]

Drone Model	W (lb)	Max Forward Speed (ft/s)	No. of Prop	Prop Diameter (in)	Standard Nozzle Model	Max Spray Flow (Gal/min)	Atomized Particle Size (μm)	Min V_{wake} per prop (ft/s)
Agras T30	169	22.97	6	20	SX11001VS	1.9	130-250	105
Agras T10	55	22.97	4	9	SX11001VS	0.5	130-250	161
Agras T16	90	22.97	6	9	XR11001VS	0.95	130-250	167
Agras T20	94	22.97	6	9	SX11001VS	0.95	130-250	171

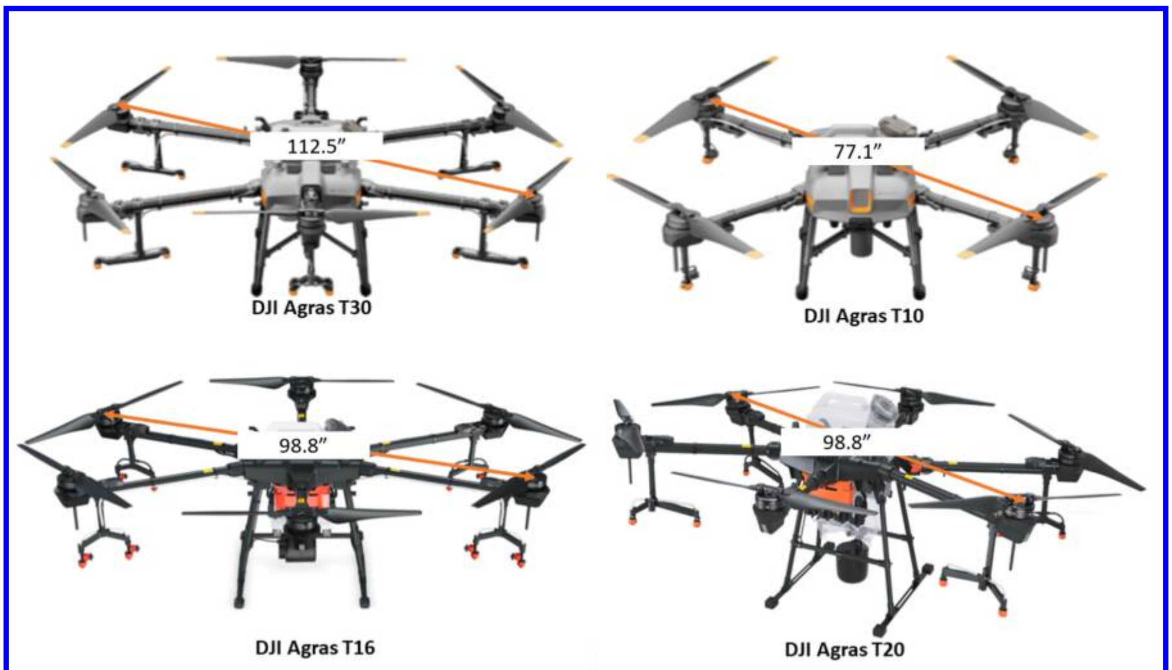

Figure 1 DJI Agridrones [6]

In the DJI drones, the spray nozzles are placed at a distance downstream of the propeller, either directly underneath the hub or at a distance inboard towards the drone center. Using actuator disk theory, the mean wake velocity can be determined by Equation 1.

$$V_e = \sqrt{\frac{2\,W/n}{\rho A} + V_0^2} \tag{1}$$

As seen in Table 1, the weight of the drones' ranges from 55 kg to 169 lbs with a maximum of 6 propellers. The minimum wake velocity (assuming negligible V_0) ranges from 30 m/s to 50 m/s. A typical droplet velocity from an agricultural nozzle range from 5 to 12 m/s [7]. Theoretically, the propeller should help increase the droplet velocity from the nozzle minimizing the drift of fine droplets. However, the wake from the propeller has localized regions of turbulence that can cause the droplets to drift out of the spray plane. The flow visualization in the wake of the propeller is shown in Figure 2.

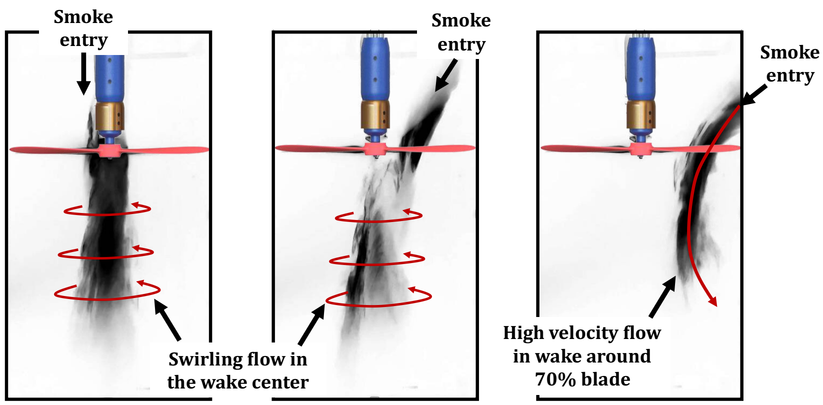

Figure 2 Smoke flow visualization in the propeller wake

A strong swirling flow can be seen in the center of the propeller wake that can cause the droplets to drift away from the spray plane. It can also be observed that the wake velocity increases from the wake center to the periphery. The presence of strong velocity gradients in the wake has the potential to increase the drift potential of the droplets and can lead to severe off-target deposition.

The United States Environmental Protection Agency (EPA) defines spray drift as the physical movement of agricultural chemicals through the air to any site other than the site intended for application [8]. Spray drift poses a significant health risk when applied in traditional ways, leading to acute/chronic illnesses in exposed humans, while also posing safety risks to the surrounding environment. While drift from agricultural chemicals has been shown to cause illness in exposed humans, it also poses a risk of killing sensitive crops, or of poisoning and killing fish [9, 10, 11, 12]. Spray drift can also lead to reduction in crop yields, deformities of crops, including reduction in plant height, fruit size, and flower production [2]. Drift in general can be caused by the method of spraying, unfavorable conditions (i.e., high wind speeds), excess or repeated use of chemicals, careless application, or volatile chemicals [2]. Of the many factors that affect drift there is still no consensus on which has the greatest impact. Bergeron et al. [13] concluded that crosswind speeds greater than 5 m/s are associated with a large risk of drift, and Gurjar et al. [14] claimed that wind speed is the main factor affecting drift. Hewitt et al. [15] claimed that that droplet size is the primary factor influencing both on and off target deposition. Irrespective of the disagreement in the most contributing factor to spray drift, there is general agreement that droplets below 200 microns or 150 microns [5, 13] in diameter have significant potential for drift and droplets under 50 microns in diameter will evaporate before reaching the target surface [16].

Many studies have been done recently examining the factors that affect spray drift related to UAVs. Areas of interest include droplet size, nozzle type and size, forward speed, flying height, and UAV design or nozzle location, among other things including environmental factors. It has been shown that both downwash and vorticity from a UAV are beneficial for deposition and penetration into the canopy. Using a series of droplet collection lines and a 4 rotor UAV, Guo et al. [17] found that stronger vorticity, which increases with forward speed, led to higher deposition quality, indicating that to a certain extent, increase in forward speed was beneficial for deposition. This finding was corroborated using a similar method with water sensitive paper by Zhan et al. [1] who claimed vorticity induced by the propellers enhanced penetration into the canopy and decreased drift. On the contrary, Ilker et al. [18] concluded that increase in forward speed led to an increase in pressure on the nozzle causing finer droplets to form, leading to an increase in drift potential, and Sheng et al [20] found the coupling of flow fields from multiple rotors causes a highly turbulent wake where vortices join and are subsequently destroyed through CFD simulations which were validated through wind tunnel testing. Sheng et al. [20] claims this flow field coupling is the main reason for uneven droplet distributions. Using a combination of agricultural dispersal models and helicopter flow field models, Teske et al. [21] demonstrated the importance of critical speed that should not be exceeded for spraying applications. Qi Liu et al. [22]

collected droplets at 10 locations downwind of a UAV placed in an open jet wind tunnel at various heights and reported that UAVs reduce the potential for spray drift when compared to fixed wing aircraft due to low flying height. However, another study conducted by the same researchers [22] found droplets deposited above the propeller plane downwind and found that nearly 5X the number of droplets adhered to the fuselage of a UAV when compared to a tractor boom, while sheng et al. [20] also found droplets recirculating above the UAV and back through the propeller indicating that UAV downwash can still lead to a high risk of drift.

As seen in literature, there are conflicting claims on the extent of spray drift from drone spraying that largely inhibits efficient drone farming. It also restricts development of safety protocols and best practices and could misinform restrictions in flight height, flight speed, wind speed, spray quality, etc. A systematic study is required to properly assess the extent of propeller wake influence in the droplet size distribution of sprays using more than one experimental methodology. In this work, we spray water using standard American Society of Agricultural and Biological Engineers (ASABE) nozzles placed at different locations in the wake of one and two propellers to characterize the changes in the droplet size distribution. We employ laser diffraction technique to determine droplet size distribution along with a gutter test to determine the overall spray volume with and without the propeller. High-speed shadowgraph technique was also employed to visualize the influence of wake on atomization of sprays. The shadowgraph images were coupled with a statistical algorithm developed by Wanstall et al. [23] to quantify and characterize full-field liquid existence along the line of sight of the spray for the different propeller-nozzle configurations tested.

III. Experimental Setup

All experiments were conducted in the University of Dayton Wind Tunnel Laboratory using the propeller-spray test stand shown in Figure 3. The test stand can hold one or two APC 17x7 propellers on two parallel linear traverses with a traversing distance of 3’2”. Pipe fittings were installed to hold standard agricultural nozzles in the wake of the propeller at varying transverse and downstream distances. Standard ASABE nozzles based on the Fritz ASABE nozzle classifications [24] were used in the study. The nozzles are supplied constant pressure using a Burcam constant pressure pump, and pressure was measured using an SSI technologies model MG1-200-A-9V-R digital pressure gauge with an accuracy of $\pm$0.25%.

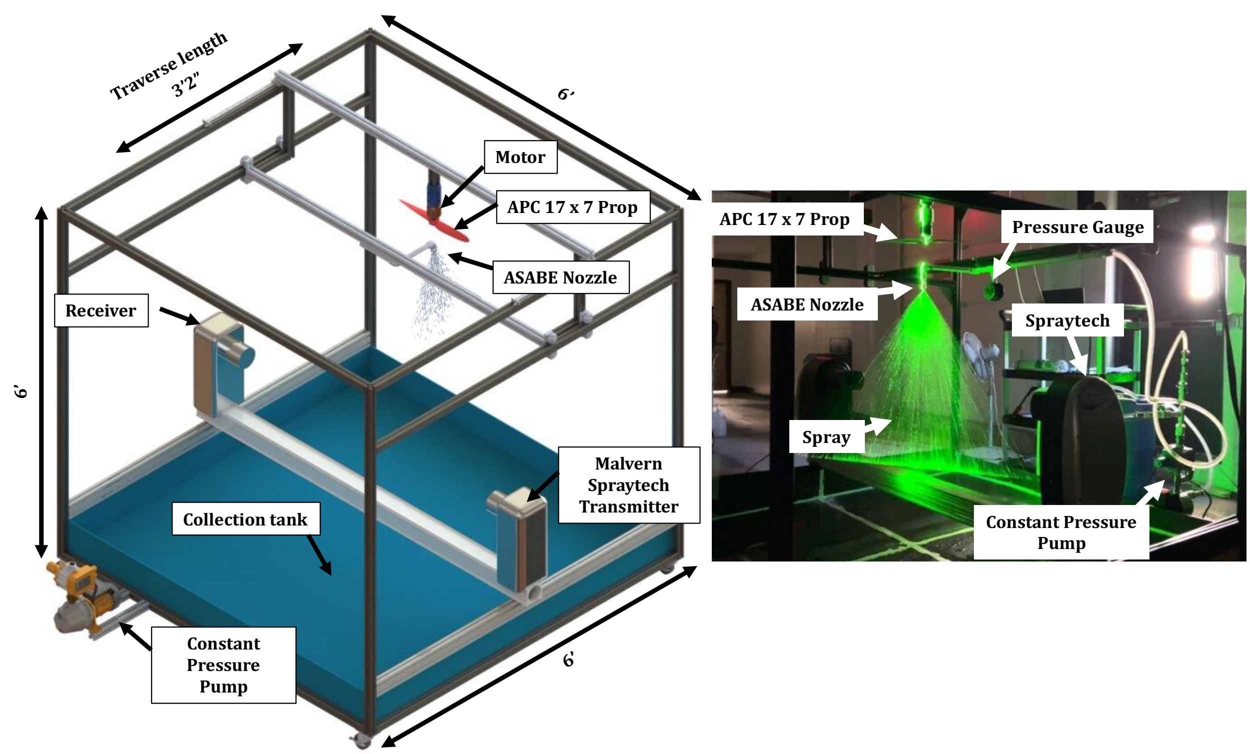

Figure 3 Schematic of the propeller-spray test stand

The nozzles are classified based on spray angle, volumetric flow rate, and droplet size ranging from very fine to extremely coarse. For example, to "11001" nozzle has a spray angle of 110-degrees with an expected output of 0.1 gallons per minute. All specification for the nozzles tested are shown below in Table 2. At the rated pressure of 21.7 psi, the reported volumetric flow rate for the 6515 nozzle is 1.15 GPM as opposed to 1.5 GPM as mentioned in the Fritz ASABE nozzle classifications [24]. All testing was conducted using water as the medium. The laser diffraction, gutter tests, and the high-speed shadowgraph experiments were conducted in the test stand and the respective methodologies are discussed in the sub-sections below.

Table 2 Specifications of the ASABE Standard Nozzles [24]

Nozzle	Fan Angle (Deg)	Volumetric Flow Rate (GPM)	Pressure rating (PSI)
11001 (very fine)	110	0.1	65.3
11003 (fine)	110	0.3	43.5
11006 (medium)	110	0.6	29
8008 (coarse)	80	0.8	36.3
6510 (very coarse)	65	1.0	29
6515 (extremely coarse)	65	1.15	21.7

A. Droplet Size Distribution Measurement using Laser Diffraction

The droplet size measurements were obtained using the Malvern Spraytec laser diffraction instrument [25] that can measure range of droplets from 2-2500μm with $\pm 1\%$ accuracy. The schematic of the nozzle locations tested underneath the propeller is shown in Figure 4.

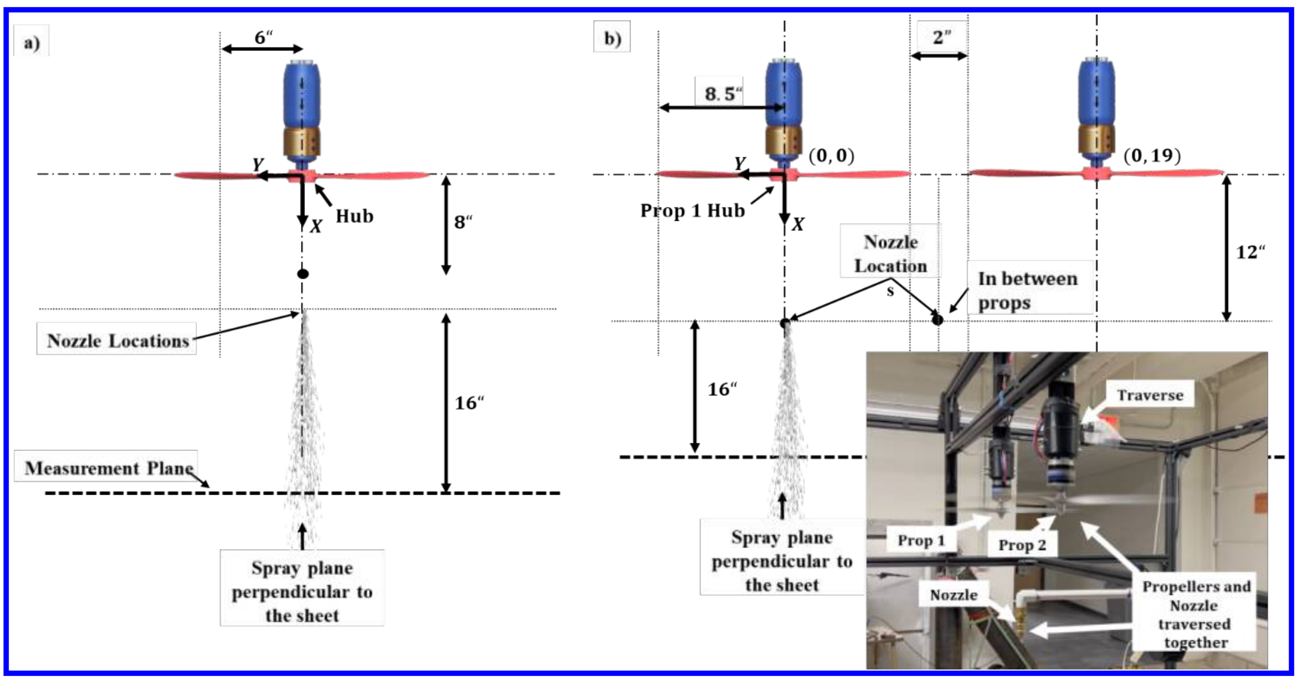

Figure 4 Schematic of propeller – nozzle configurations tested with a) 1 propeller b) 2 propellers

All six nozzles were tested underneath a propeller at two different locations, one below the hub and the other at 70% of the radius of the propeller where the blade loading is at its highest. The distance between the nozzle and the propeller plane was fixed at 8". The nozzles were tested without propeller running, and at 3900 and 5900 propeller RPMs. The plane or the fan of the spray is oriented into the paper as it was traversed through the measurement zone of the laser diffraction instrument. The traverse motion and the droplet size data acquisition was synchronized in time. A second propeller, Prop 2, was added adjacent to Prop 1 to test the influence of propeller interactions in droplet size distributions. In the two-propeller configuration, the nozzles were placed at two different locations, one below the hub of Prop 1 and the other at the midpoint between the two propellers. The distance between the nozzle and the propeller was fixed at 12". Similar to the single propeller configuration, the two-propeller configuration along with the nozzle were traversed across the measurement zone of the laser diffraction instrument. The test matrix to obtain the droplet size distribution data in the wake of one and two-propeller configurations is shown in Table 3.

Table 3 Test matrix to obtain droplet size distribution data in the wake of propellers

Nozzles	No. of Props	Y - Nozzle Location	X – Nozzle Location	Prop RPM
11001, 11003, 11006, 8008, 6510, 6515	1	Hub, 70% R_{Prop}	8" below Prop	0, 3900, 5900
11001, 11003, 11006, 8008, 6510, 6515	2	Prop 1 Hub, 2 Prop Center	12" below Prop 1	4900

B. Spray Volume Measurement using Gutter Test

The droplet size distribution measurements from the laser diffraction instrument necessitated measuring the overall volume of the spray from the nozzles by placing gutters underneath the nozzles and measuring the volume of spray collected in a time interval. A total of 15 gutters were placed below the spray plane oriented along the fan of the spray as shown in Figure 5. The volume of spray collected in each gutter was measured using a measuring cup. Three tests were conducted for each nozzle for data repeatability. The test matrix for the gutter test is shown in Table 4.

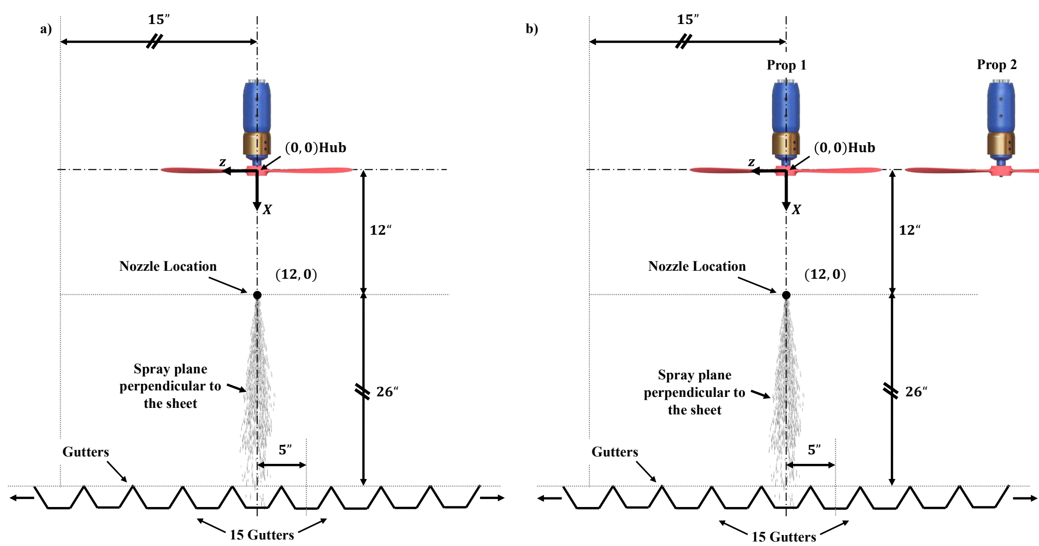

Figure 5 Gutter test conducted with a) 1 propeller b) 2 propeller configurations

Table 4 Test matrix for the gutter test

Nozzle	Spray Time (s)	No. of Prop	X - Nozzle Location	Y-Nozzle Location	Prop RPM
11001 (very fine)	30	1, 2	Prop 1 Hub	12" below Prop 1	0, 4900
11003 (fine)	30				
11006 (medium)	25				
8008 (coarse)	20				
6510 (very coarse)	15				
6515 (extremely coarse)	15				

C. Spray Visualization using High Speed Shadowgraph

High speed shadowgraph was conducted to visualize the effect of propeller wake on spray droplets in different configurations. The light source consisted of several high-power LEDs placed behind an 8' x 4' long Tuffak® Lumen XT light diffuser that was used as the background. A Phantom VEO 1310L camera with a 1280 x 960 pixel resolution fitted with a Nikon 105 mm lens was used to capture shadowgraph images of the spray. An exposure time of 6 μs was found to be sufficient to avoid motion blur. The schematic of the high-speed shadowgraph setup is shown in Figure 6a. The resultant shadowgraph image is shown in Figure 6b. The field of view (FOV) was trimmed to 1' x 1'3" to focus on the spray. The test matrix for the shadowgraph experiments is shown in Table 5. All six nozzles were tested underneath a propeller at two different locations, one below the hub and the other at 70% of the radius of the propeller. In the two-propeller configuration, the nozzles were placed at two different locations, one below the hub of Prop 1 and the other at midpoint between the two propellers. The distance between the nozzle and the propeller for both cases was fixed at 12".

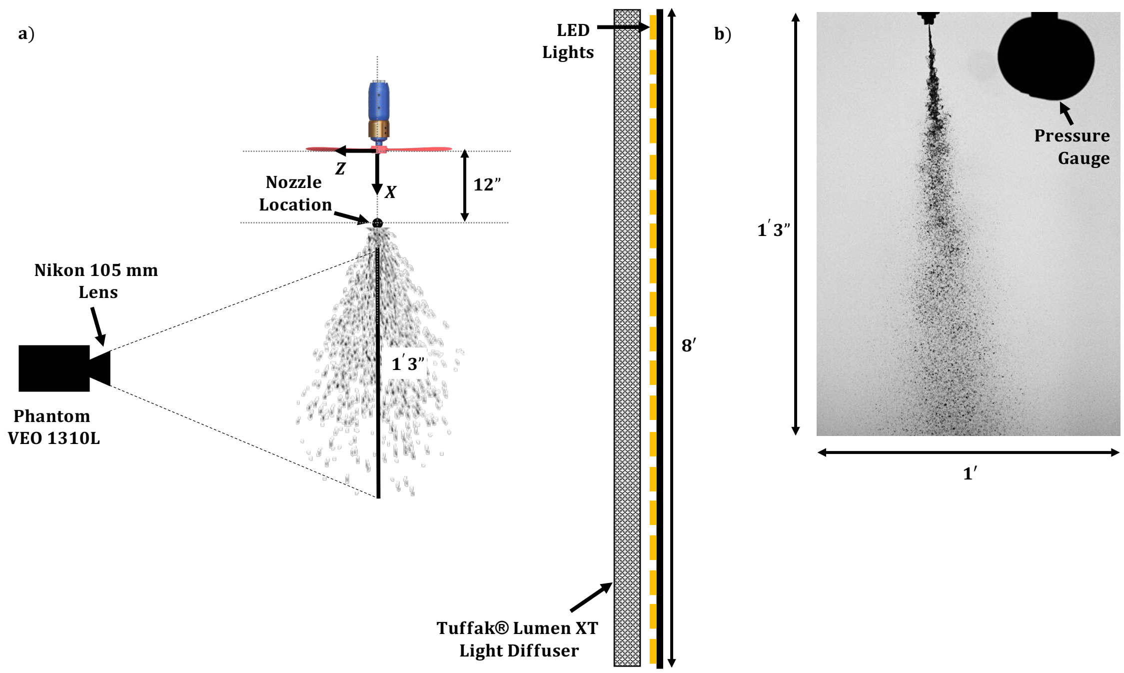

Figure 6 a) Schematic of the high-speed shadowgraph test setup b) Resultant shadowgraph image showing the FOV size

Table 5 Test matrix for the high-speed shadowgraph experiments

Nozzles	No. of Props	Y - Nozzle Location	X – Nozzle Location	Prop RPM
11001, 11003, 11006, 8008, 6510, 6515	1	Hub, 70% R_{Prop}	12" below Prop	4900
11001, 11003, 11006, 8008, 6510, 6515	2	Prop 1 Hub, 2 Prop Center	12" below Prop 1	4900

These shadowgraph images were coupled with an extinction-based statistical algorithm to create probability contours based on the presence of droplets in the field of view. The algorithm first compiles intensity values for each pixel location in the images. A reference or threshold intensity as the percentage of the background intensity is then determined through a sensitivity analysis that determines the last axial location where liquid is present 100% of the time. The threshold values sweep a range of 50-99% of the background value for each pixel. Next, each pixel in the respective images is binarized to a value of 1 or 0 assigning 1 to pixels where liquid is present. The binary values are then summed for each pixel. The sums are then divided by the total number of images yielding the probability of liquid existence at each pixel in the image. The development of the algorithm and the validation is discussed in detail in [23].

IV. Results

The results of the various experiments will be discussed in sub-sections. First, the droplet size distribution results will be discussed for all the cases tested followed by the gutter test results. Then the gutter test and the droplet size distribution data will be compared with the results from high-speed shadowgraph and liquid probability for selected cases.

A. Droplet Size Distribution Results

A.1 Droplet Sensitivity to Propeller RPM and Nozzle Distance from Propeller

The droplet size distribution data, from the single propeller data for all six nozzles placed underneath propeller hub are shown in Figure 7 for various propeller RPMs. The baseline standalone nozzle cases are shown in black. The effect of the propeller on the spray is extremely evident. In all the nozzle cases, the droplet size distribution shifts towards the right of the graph indicating larger droplet sizes in the spray. The different RPM cases show measurable differences for the finer nozzles (11001 and 11003) but the differences between the RPM cases diminish for medium (11006 and 8008) to coarser nozzles (6510 and 6515 nozzle). This indicates that the fine spray nozzles are more sensitive to propeller RPM whereas the RPM dependency significantly decreases for medium to coarse nozzles. There are also no measurable differences when the nozzle is placed 8" or 12" below the hub except in the case of the 11001, very fine, nozzle. Similar to the RPM sensitivity, the droplet size distribution is sensitive to the nozzle placement underneath the propeller only for the finer nozzles. The medium to coarse nozzles shows very little or negligible differences to nozzle location underneath the propeller plane.

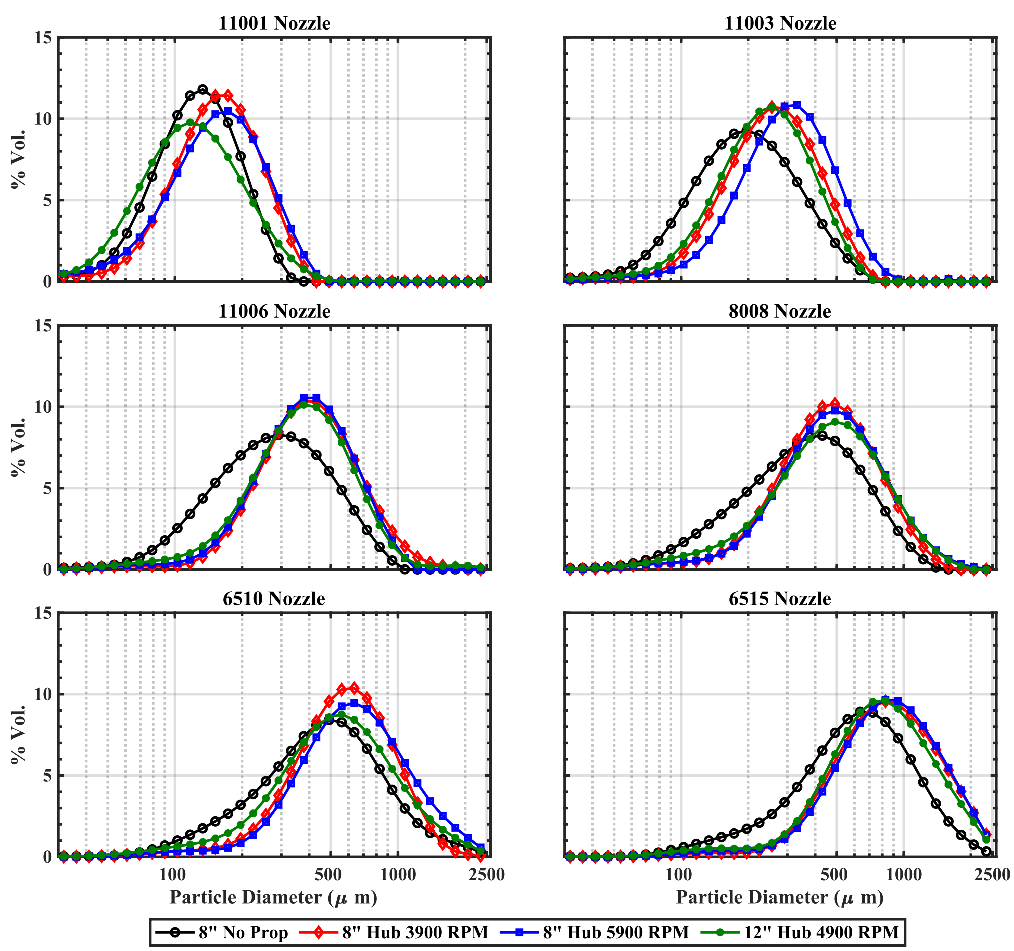

Figure 7 Droplet size distribution data for 1 prop case for all six nozzles at various propeller RPM and distance from the propeller plane

A.2 Droplet Sensitivity to Nozzle Placement in the Propeller Wake with 1 Propeller

The droplet size distribution data from a single propeller for all six nozzles placed at the hub and at the 70% R_{Prop} are shown in Figure 8 for different RPM cases. Like the trends seen in Figure 7, all cases show the shift to the right of the curve indicating a change to larger droplet sizes. The cases where the nozzles are placed at the 70% R_{Prop} is shown in dotted lines. For almost all cases, the droplet sizes from the 70% R_{Prop} location falls in-between the no-propeller cases and the hub case. This indicates that the droplet sizes are sensitive to the nozzle location in the wake. As the nozzle is traversed away from the hub of the propeller, the droplet size distribution more closely resembles the standalone propeller as the propeller wake influence decreases.

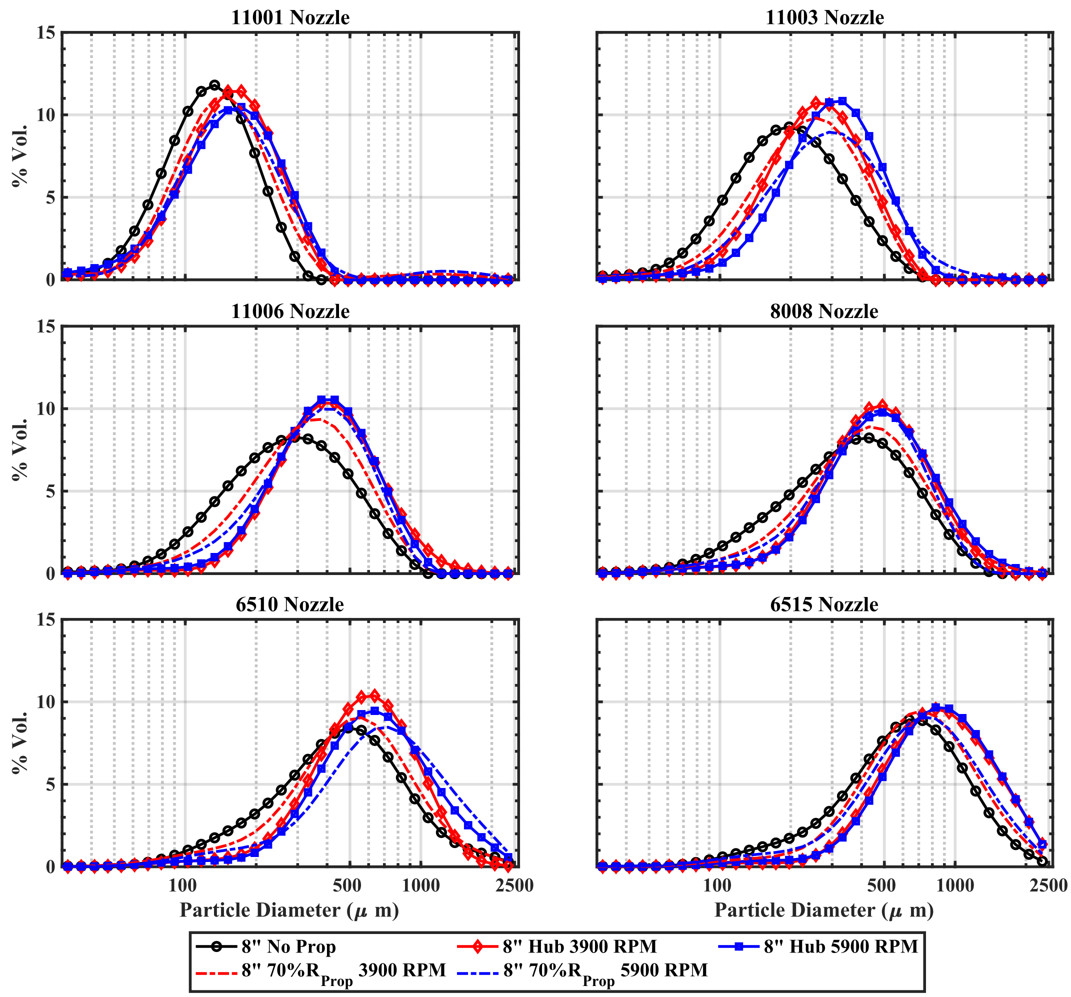

Figure 8 Droplet size distribution data for 1 prop case for all six nozzles placed at different locations in the propeller wake (Propeller hub and 70% R_{Prop})

A.3 Droplet Sensitivity to Number of Propellers

The changes in the droplet size distribution results with different number of propellers are shown in Figure 9 where the nozzles are placed 12" below the propeller hub. For the case with two propellers, the nozzles were placed at 12" below the Prop 1 hub as indicated in Figure 4. For the large droplets corresponding to each of the nozzle (descending end of the bell curve), the second propeller doesn't influence the distribution significantly, except for the extremely coarse 6515 nozzle where the second propeller resembles the no-propeller case. The volume of the spray containing smaller droplets does show measurable sensitivity to the number of propellers as annotated in Figure 9. The single propeller case yields a lower volume of finer droplets when compared to the two-propeller case for almost all nozzles.

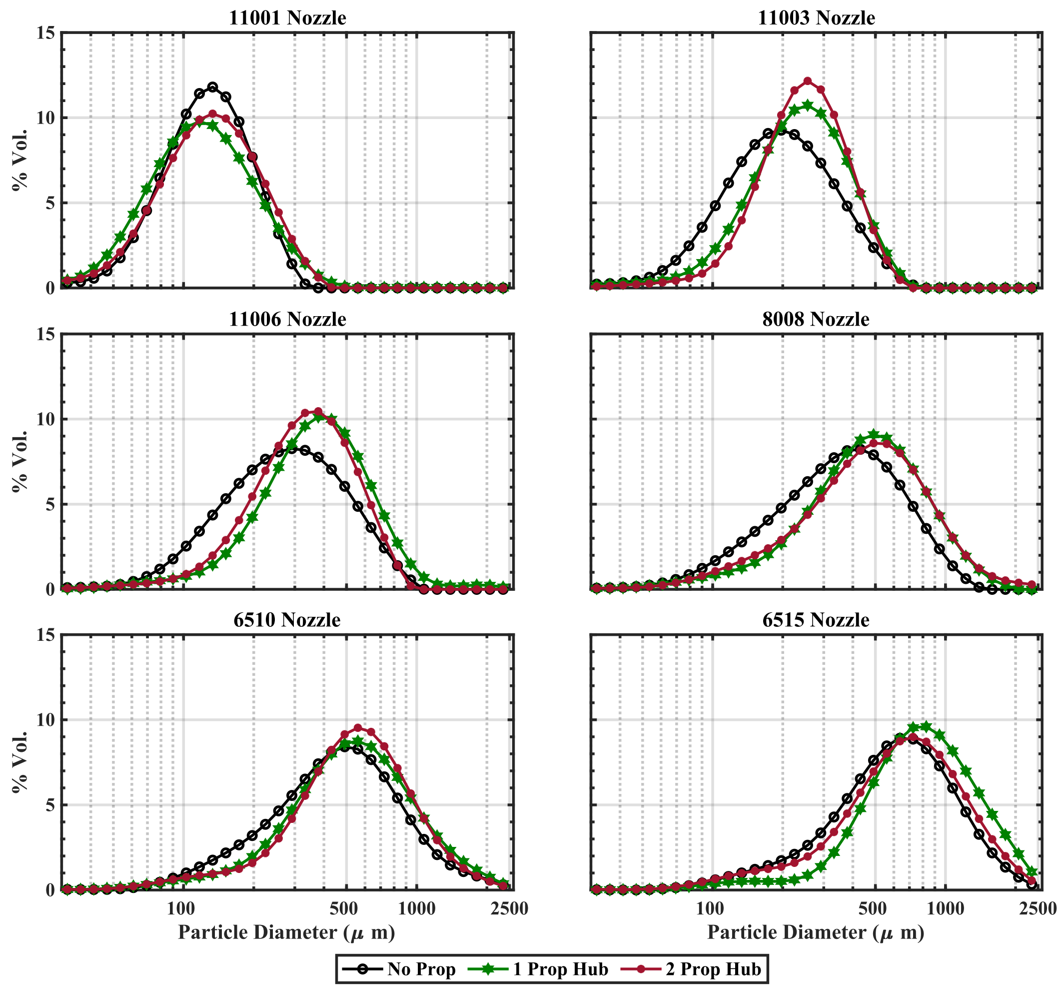

Figure 9 Droplet size distribution data for all six nozzles placed underneath Prop 1 hub with different number of propellers.

A.4 Droplet Sensitivity to Nozzle Placement in the Propeller Wake with 2 Propellers

Similar to the sensitivity of the droplet sizes to the nozzle placement with 1 propeller shown in Figure 8, the droplet size sensitivity to the nozzle placement in the wake of 2 propellers is shown in Figure 10. In the first location, the nozzles were placed directly underneath the hub of Prop 1 and in the second location, the nozzles were placed in-between the two propellers. In all the cases shown in Figure 10, the nozzles were placed at 12" below the propeller plane as shown in Figure 4. Like the number of propeller sensitivity results, the placement of the nozzles at different locations shows similar behavior. The larger end of the droplet size curves shows no measurable differences between the nozzle locations. However, the finer droplets from all six nozzles show slightly different behavior based on the nozzle location. Placing the nozzles directly underneath the Prop 1 hub seems to yield a lower volume of finer droplets when compared to the nozzle placement at the center of the two propellers for almost all nozzles.

A.5 Summary of Droplet Size Distribution Sensitivities

The droplet size distribution results indicate that the influence of the propeller wake shifts the droplet size distribution to larger droplet sizes for all six standard nozzles. The medium to extremely coarse nozzles are less sensitive to propeller RPM and the location of the nozzle from the propeller plane. The droplet size distributions show measurable sensitivity to the presence of the second propeller, however the placement of the nozzles in-between the two propellers yield similar droplet distributions to that of the nozzle place underneath the hub of Prop 1. Overall, the largest increase in droplet size is seen when the nozzle is placed directly underneath the hub of the propeller, either with one propeller or two propellers. In general, increase in the overall droplet size distribution correlates to fewer driftable droplets meaning that drone spraying can lead to a decrease in spray drift. Several claims from the literature on the reduction in drift using drone are based on similar droplet size distribution studies. However, as indicated in Figure 2, the wake of the propeller is extremely turbulent and is largely three-dimensional and the three-dimensional influence of the wake on the droplets needs to be considered.

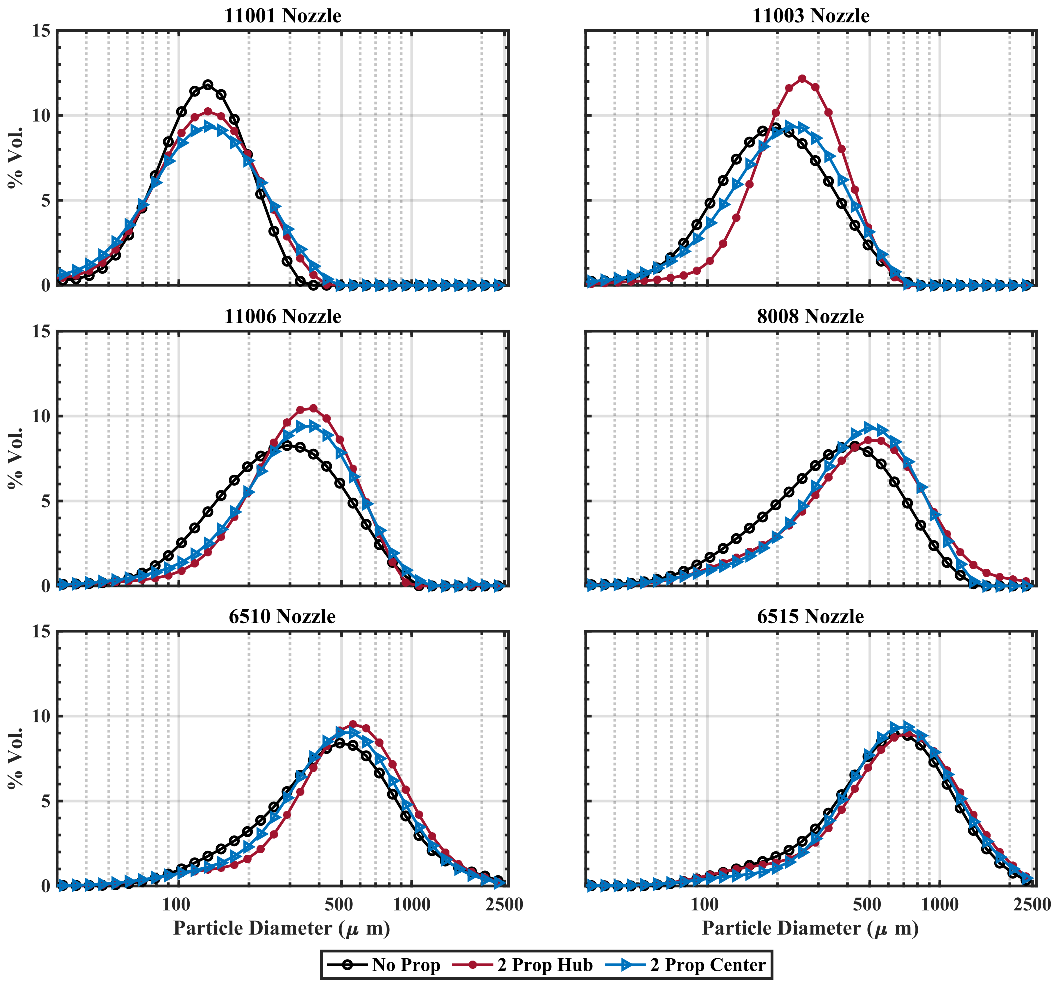

Figure 10 Droplet size distribution data for all six nozzles placed underneath Prop 1 hub and in-between two propellers

While the droplet size distributions measured using a point-wise laser diffraction measurements show a shift in the distribution towards larger droplets, simultaneous flow visualization conducted using a pair of 300mW, 532 nm laser shows an interesting dichotomy between the droplet size results and the overall droplet dynamics. As seen in Figure 11a, it is extremely evident that droplets exist outside of the test stand, and it is highly unlikely that all of them passed through the line-of-sight laser diffraction measurement zone, even when the nozzle-propeller system is traversed together. It would appear that the swirl component and the highly three-dimensional nature of the propeller wake (as shown in Figure 2) carries the finer droplets outside the spray plane, and therefore, a lesser total spray volume is measured by the laser diffraction instrument when compared to the no-propeller case. Figure 11a also shows the presence of a large recirculation region that forms only when the propellers are activated sending the fine droplets nearly 6' away from the spray origin and towards the ceiling. The suction effect of the propeller brings these droplets back into the propeller plane and gets reintroduced to the spray measurement zone as seen in Figure 11b. This pattern of large-scale recirculation is consistent with research conducted by Qi et al. [21, 22] that showed droplets were collected above the propeller plane and adhered to the UAV fuselage. The flow visualization of droplets casts doubts upon the validity of using droplet size distribution results to make decisions related to drift potential of drone spraying.

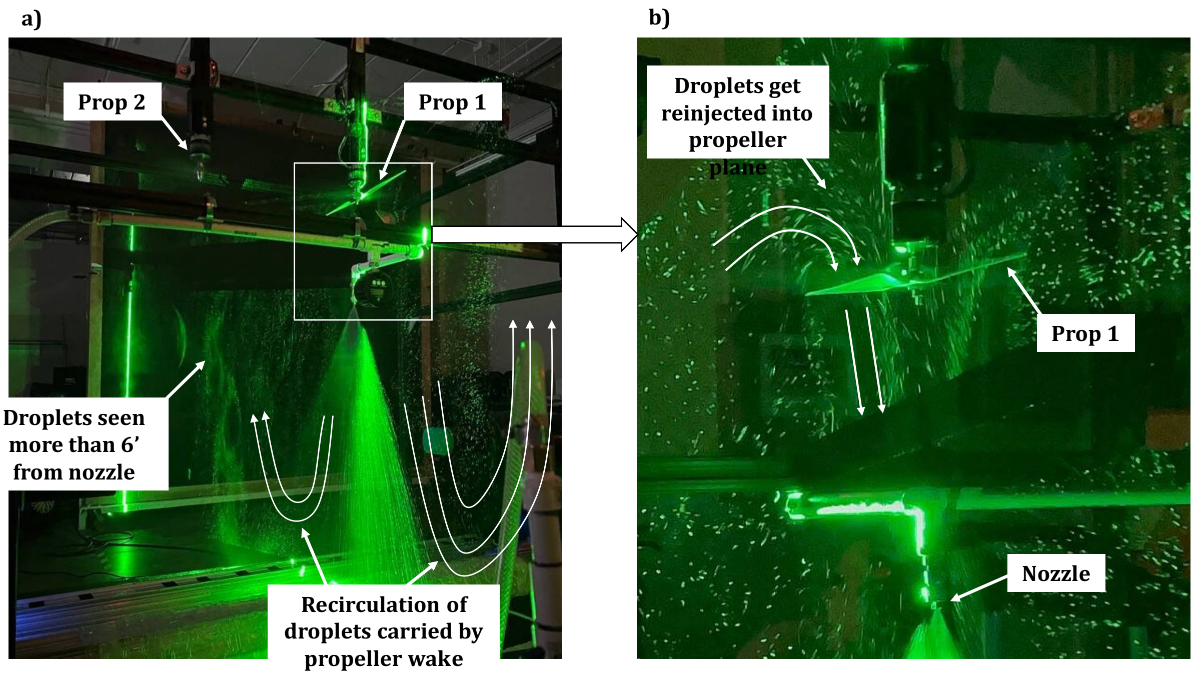

Figure 11 Flow Visualization for a nozzle beneath the hub for a two-propeller configuration

B. Spray Volume Measurement using Gutter Test

To quantify the volume of spray never measured by the laser diffraction instrument, gutter tests were conducted to collect the spray deposition using the test setup shown in Figure 5. The results of the gutter tests are shown in Figure 12. All volume measurements were normalized with respect to the no propeller case for all the nozzles tested.

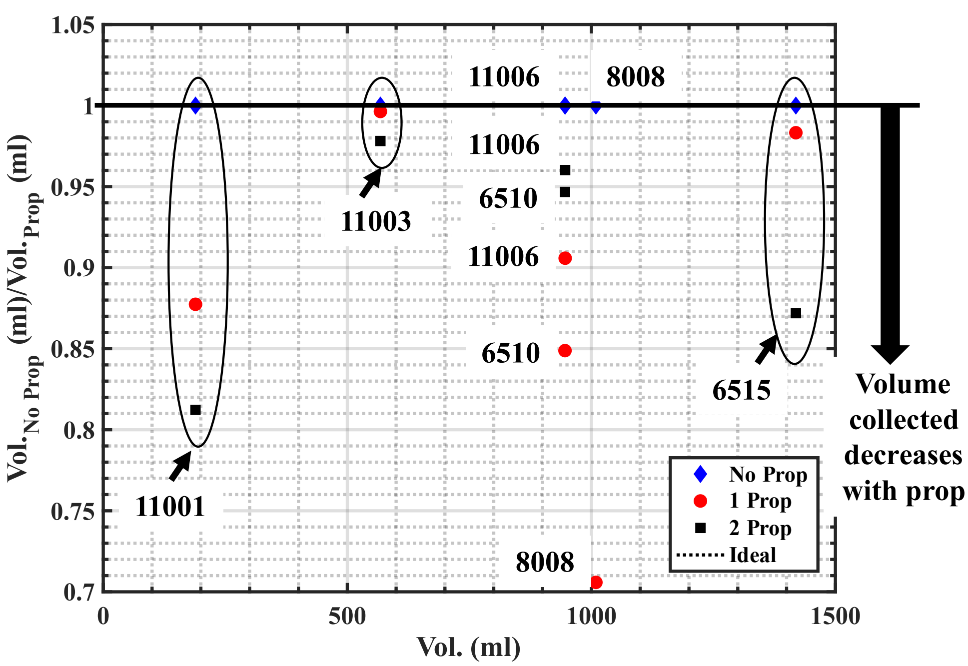

Figure 12 Normalized Volume collection averages for all 6 nozzles placed 12” beneath the hub of 1 and 2 propellers

The trends in Figure 12 corroborates the flow visualization results shown in Figure 11 where the smaller droplets are being carried away from the spray plume, and never reaching the laser diffraction measurement zone. Across all 6 nozzles, the introduction of the propellers led to a significant decrease in overall volume collected over the same time interval. The data shown in Figure 12 is the average of three trials conducted for a given case. While there are inherent uncertainties associated with the measurement due to the crudeness of such a test, there exists a global trend where volumes in the order of 10%-20% is never collected due to the influence of the propeller wake. It was determined the most volume lost is when the nozzle was directly underneath the Prop 1 hub in both 1 and 2 propeller configurations. While the droplet size data shows that nozzle placement under the hub minimizes the drift potential (due to larger droplet sizes), flow visualization and the gutter test data shows that it is the worst place for the nozzle to be as the high swirl component of the propeller ejects the finer droplets out of the spray plane and the measurement zone significantly increasing the drift potential. A more rigorous approach was taken to quantify the spray volume that is lost due to the propeller influence through extinction-based imaging, i.e., shadowgraph images, coupled with a robust post processing algorithm adapted from [23] discussed in the following section.

C. Spray Visualization using High Speed Shadowgraph

The results from the high-speed shadowgraph were used to quantify the direct changes in the spray for different nozzles due to the influence of the propeller and nozzle locations in the wake of the propeller. Two main parameters were used to quantify the propeller influence on the spray: Root Mean Square (RMS) of the droplet intensity and liquid probability (P(Liquid)). The RMS is calculated using Equation 2.

$$RMS = \sqrt{\frac{\sum_{i=1}^{n}\left(I_{Instant_i} - I_{Mean}\right)^2}{n}} \tag{2}$$

where $I_{Instant}$ is the instantaneous intensity of the pixel in a given time t and I_{Mean} is the time-averaged intensity of the same pixel. Equation 2 was used to calculate the RMS(Intensity) in each pixel of the shadowgraph image. For each case, 1500 images were taken with a sampling rate of 300 frames per second. Images were not collected during the startup transience of the spray and the propeller. The data was collected only after the spray achieved a steady state and the propeller reached a steady state RPM of 4900. This means that the total volume being emitted from a given nozzle across the 5 cases shown in this section is almost identical with a nearly negligible difference due to the uncertainty of the pressure sensor. Figures 13-15 below, all show a reduction in liquid probability as downwind distance from the nozzle increases. The introduction of the propeller wake causes smaller droplets to form upon atomization of the spray. Larger droplets remain in the center of the spray plume, while smaller droplets are carried away, and extremely fine particles are removed from the test stand altogether as discussed in the previous sections. It is especially evident for the 11006 nozzle, the introduction of propeller wake creates droplets that are smaller than the pixel size and cannot be recognized by the algorithm. These fine droplets are likely being removed from the spray plume entirely due to the tangential component of the propeller wake velocity.

The RMS and the P(Liquid) results are shown for three nozzles, the very coarse 6510 nozzle, the coarse 8008 nozzle, and the medium 11006 nozzle. For each nozzle case, the no propeller, nozzle at the propeller hub, nozzle at 70% R_{Prop}, nozzle at Prop 1 Hub with two propellers, and nozzle at the center of two propeller cases were compared against each other. The shadowgraph images are presented in the first row, followed by the RMS, and then the liquid probability.

The results for the very coarse 6510 nozzle are shown in Figure 13. While the shadowgraph images show minimal visible differences amongst the different cases, a slight change in the inclination angle of the spray is observed for the hub cases. This is due significant oscillations in the spray pattern being induced by the propeller when the nozzle is placed directly underneath the hub. The RMS contours show the fluctuations in the spray increasing along the spray axis. The RMS region indicates the atomization zone where the liquid sheet breaks into ligaments and the ligaments further breakup into droplets. This process is called atomization. Due to the influence of the propeller, the high RMS region or the atomization region is pushed slightly downstream of the nozzle when compared to the no-propeller case indicating that the propeller wake has a strong effect in the atomization process of the spray.

The dark red regions in the liquid probability contours represents 100% liquid probability. As the spray widens, the probability of finding droplets along the spray edges goes to zero. The 10% liquid probability is shown as white contour lines for all cases shown in Figure 13 to highlight the boundaries of the spray. For the no-propeller case, a dark red region can be seen indicating the probability of finding droplets is extremely high. However, with the influence of the propeller, especially underneath the hub cases, the liquid probability decreases by almost 40% at 12” inches downwind from the nozzle when compared to the no-propeller case. This corroborates the gutter test results shown in Figure 12 where less total volume was measured in the spray with the propellers activated. The probability contours show that when the nozzles are placed at 70% R_{Prop} and at the center of the two propellers, the decrease is liquid probability is not as drastic as the hub cases. The liquid probability contours conclusively show that the droplet size distribution results are biased, and the volume of the spray being measured is not the same when the propellers are activated. The reason why larger droplets were measured in the hub cases using the line-of-sight laser diffraction measurement was because almost 10-20% of the total volume emitted is carried away by the propeller wake.

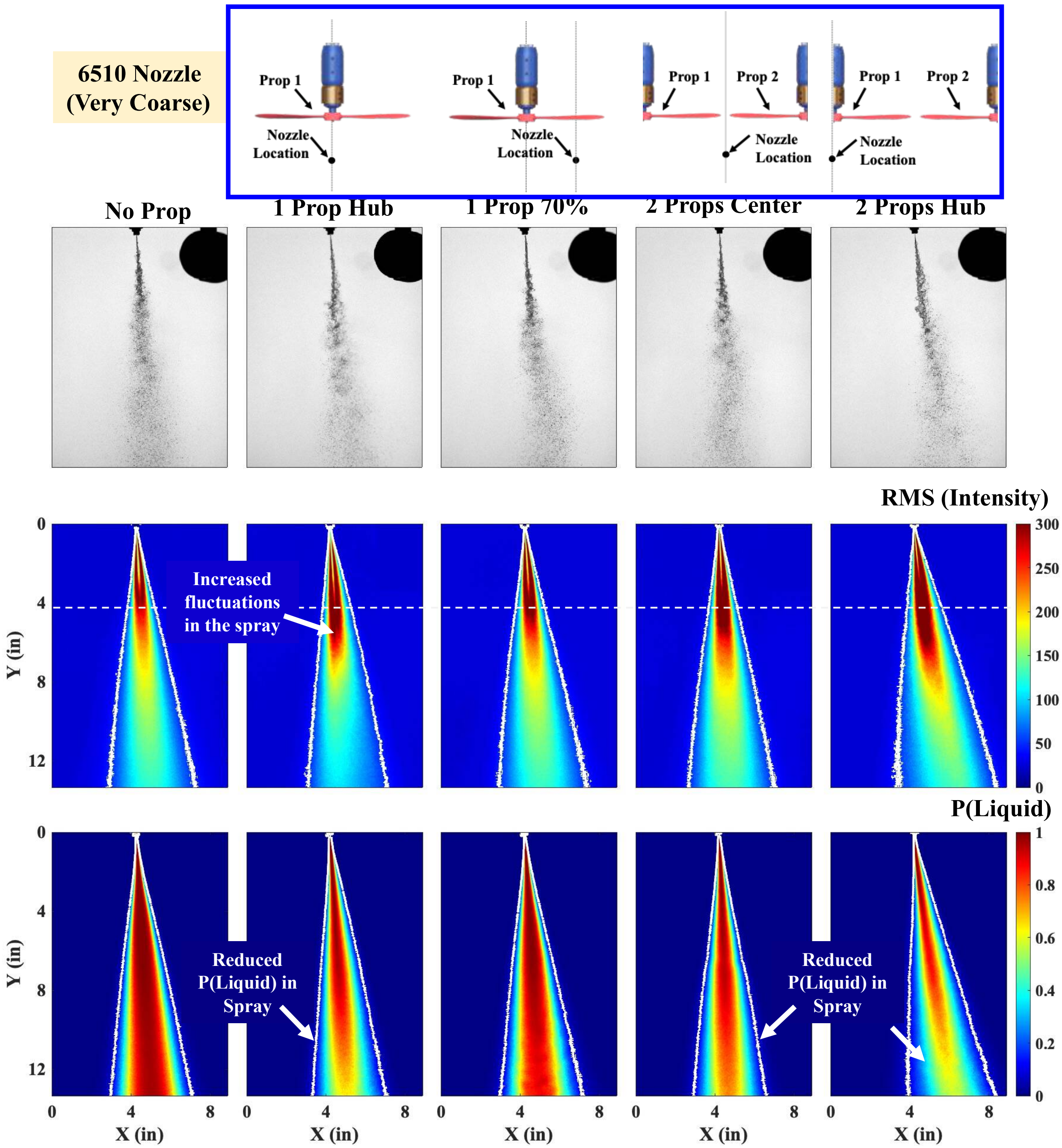

Figure 13 Shadowgraph images, RMS contours, and probability contours for a 6510 nozzle at 12" below the propeller plane

The shadowgraph images, RMS, and the liquid probability contours are shown for the coarse 8008 nozzle in Figure 14. Similar to the shadowgraph images seen in Figure 13, the extent of the spray inclination is increased for the 8008 nozzle when the nozzle is placed directly underneath Prop 1 hub for both propeller configurations. The inclination of the spray plane represents the instantenous shadowgraph image during the oscillations induced by the propeller on the spray. The increased oscillations caused an increase in spray width as seen in the RMS contours. Similar to the 6510 case, the atomization process is pushed further downstream from the nozzle for all propeller cases when compared to the no propeller cases. The changes in the atomization zone is evident in the hub cases. The probability contours shows a signficant reduction in the liquid probability, up to 60-80% at 12" from the nozzle in the hub cases when compared to the no-propeller case. While reduction is not as drastic for the $70\% R_{Prop}$ and the 2 prop center case, the liquid probabilty is still reduced when compared to the no-propeller case.

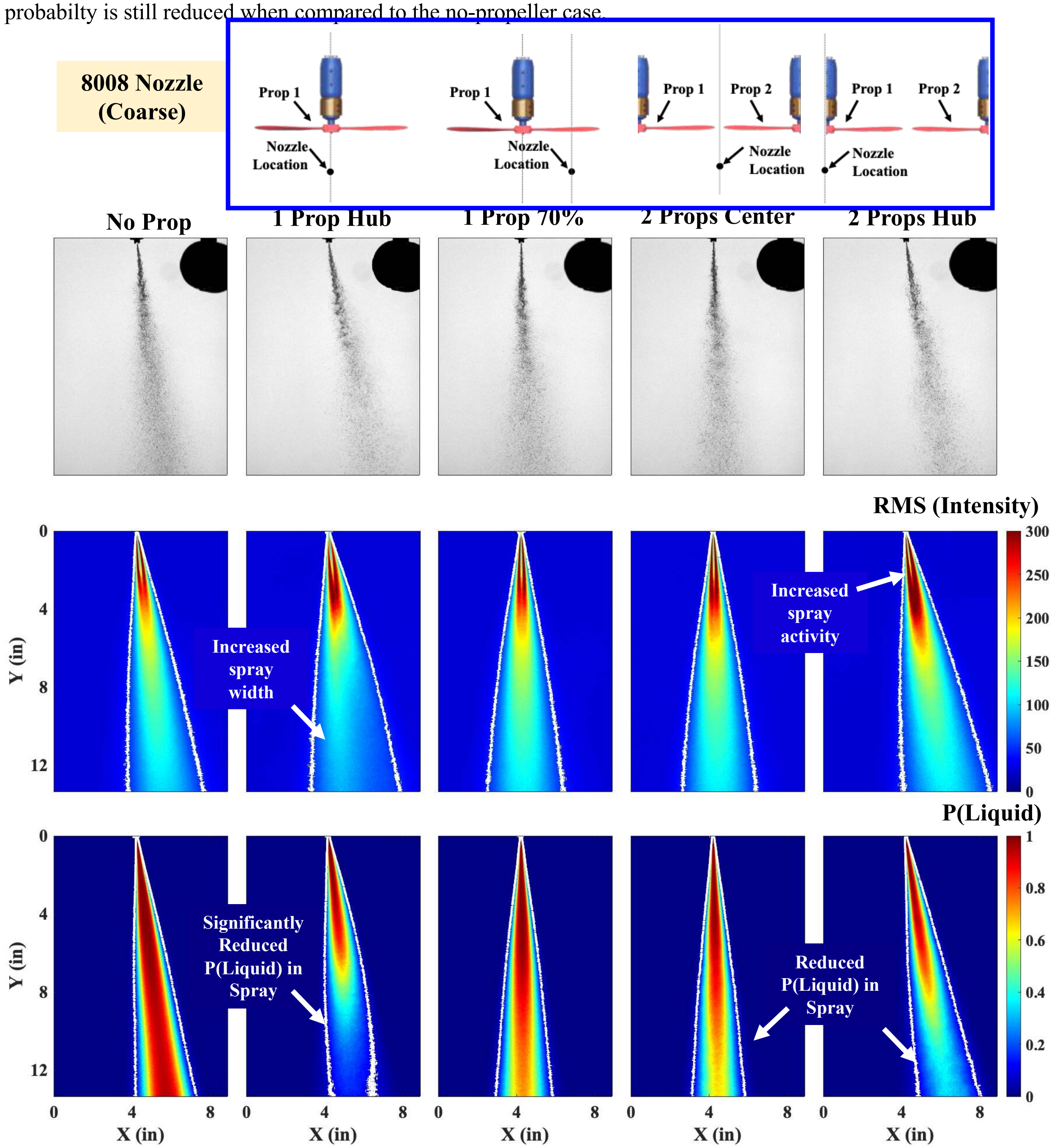

Figure 14 Shadowgraph Images, RMS contours, and Probability contours for a 8008 nozzle at multiple locations 12" downwind of the propeller plane

Finally, a similar set of contours are shown for the medium, 11006 nozzle in Figure 15. As the overall droplet size reduces, the effect of propeller on the spray increases and the propeller starts to significantly affect the spray characteristics. Since the atomization of the spray happens much earlier in the case of 11006 nozzle, the high RMS regions of the spray are concentrated closer to the nozzle when compared to the other two nozzles discussed earlier. However, similar effects can be seen where the atomization is pushed further downstream of the nozzle with the influence of the propeller. The probability contours show the biggest differences due to propeller influence for the 11006 nozzle when compared to the other two nozzles. Even at 8” from the nozzle, the liquid probability drops to nearly 0% when the nozzle is placed underneath the propeller hub. The laser diffraction measurement was done at 16” below the nozzle, where there is signifcant drop in the liquid probability for all propeller cases when compared to the no-propeller case. This once again conclusively proves that the droplet size measurement for the sprays under the propeller influence is biased and should not be used as the lone metric to assess the drift potential of the spray.

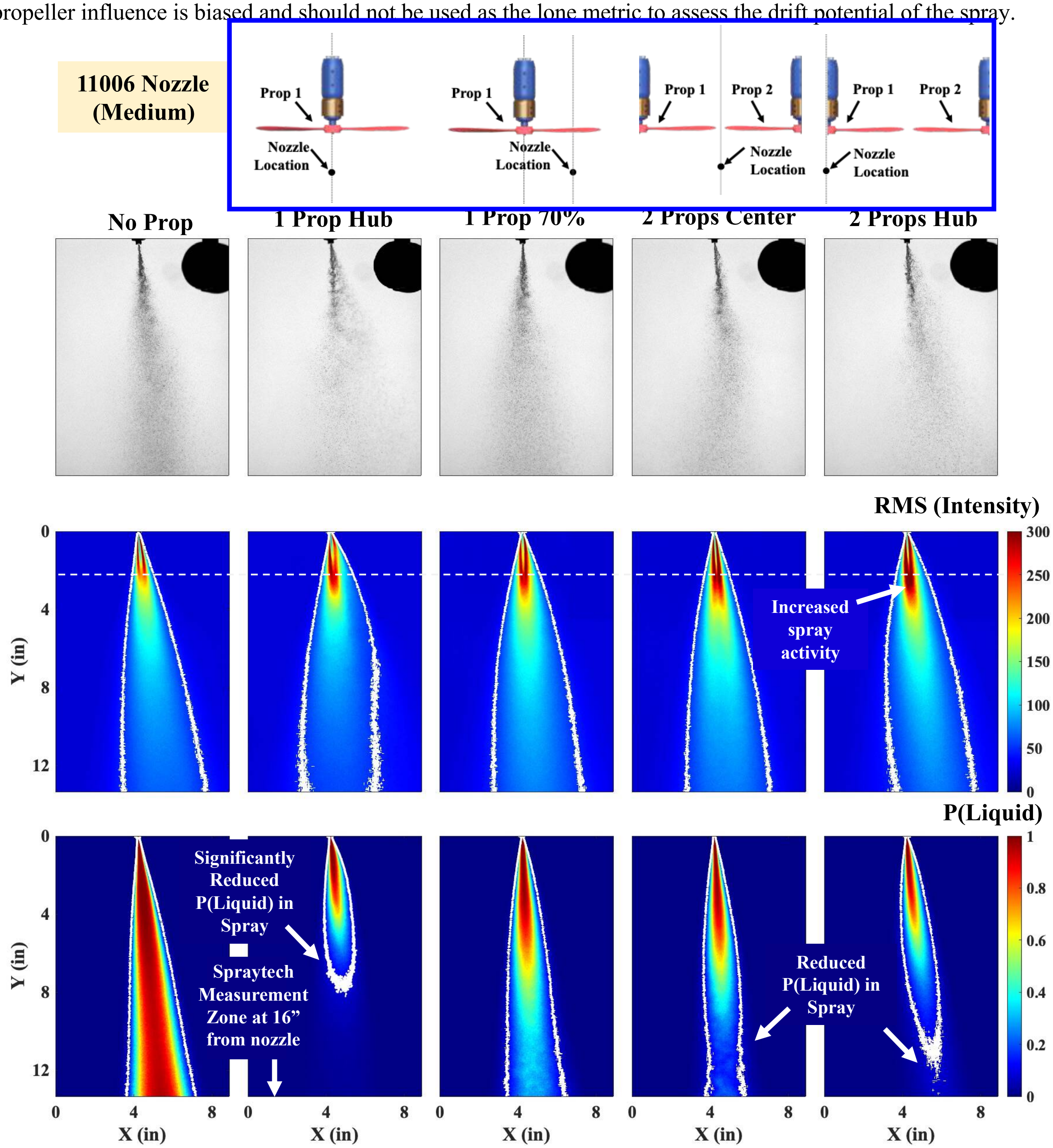

Figure 15 Shadowgraph Images, RMS contours, and Probability contours for a 11006 nozzle at multiple locations 12” downwind of the propeller plane

V. Conclusions

The influence of the propeller wake on sprays from standard ASABE nozzles was investigated using multiple experimental techniques such as laser diffraction, flow visualization, gutter volume collection, and high-speed shadowgraph. The droplet size measurements indicated that propeller influence is beneficial to the spray causing an increase in the overall droplet size, hence reducing the drift potential. However, the flow visualization results, the gutter volume collection, and the quantitative analysis from high-speed shadowgraph images indicated that the droplet size measurement using a line-of-sight laser diffraction measurement is highly biased as the laser diffraction instrument doesn't capture the entire volume of the spray when compared to the no-propeller case. As the smallest droplets are entrained in the propeller wake, the laser diffraction instrument does not detect them. As such, the overall measurement is skewed towar ds droplet population with higher mean droplet diameter.

Contrary to the droplet size measurements, the gutter test results and the flow visualization showed increased drift when the nozzle is placed directly underneath the propeller hub. The fine droplets were seen to be entrained in the highly three-dimensional nature of the propeller wake and is seen to drift more than 6' from the test location. For all nozzle cases, the total volume collected using gutters with the propeller were less than the no-propeller case by as much as 20%. The influence of the propeller on the spray was further quantified using high-speed shadowgraph images. The shadowgraph images and the RMS contours of the spray indicated that the propeller delays the atomization process and induces oscillations which significantly affects the overall liquid probability.

The liquid probability contours indicate significant reduction in liquid probability from 60 – 90% on an average at 12" below the nozzle while the droplet sizes are measured 16" below the nozzle. The flow visualization and quantification results proves to be a cautionary tale to avoid using droplet size measurement as the sole metric to assess the drift potential of drone spraying. The propeller wake effects on the droplets are much more profound and cannot be quantified using a single line-of-sight measurement technique. A secondary visualization or quantification is necessary to evaluate risk of drift from drones.

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